

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.3970847545283422, median 0.34758475820601187, std: 0.2633694193676844
Gyroscope error (imu0): mean 0.9776786739752499, median 0.7738267264814671, std: 0.7456205897007433
Accelerometer error (imu0): mean 0.6080908590964011, median 0.22553799407844524, std: 1.5505306358615136

Residuals

Reprojection error (cam0) [px]: mean 0.3970847545283422, median 0.34758475820601187, std: 0.2633694193676844
Gyroscope error (imu0) [rad/s]: mean 0.07212586255643928, median 0.05708718171151859, std: 0.05500634268040548
Accelerometer error (imu0) [m/s^2]: mean 0.011478310943141572, median 0.00425725068351079, std: 0.0292677853959716

Transformation (cam0):

T_ci: (imu0 to cam0):

[[0.02407142 -0.99943877 -0.02329612 0.00955672]
[-0.99970477 -0.02398768 -0.00386726 0.01403197]
[0.00330627 0.02338234 -0.99972113 -0.01287481]
[0. 0. 0. 1.]]

T_ic: (cam0 to imu0):

[[0.02407142 -0.99970477 0.00330627 0.01384035]
[-0.99943877 -0.02398768 0.02338234 0.01018899]
[-0.02329612 -0.00386726 -0.99972113 -0.01259432]
[0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)

-0.2617585279030665

Gravity vector in target coords: [m/s^2]

[-0.28866103 -9.80229814 0.00698546]

Calibration configuration

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cam0

Camera model: pinhole

Focal length: [922.531465350295, 925.020034192637]

Principal point: [593.0886843367651, 383.539414452546]

Distortion model: equidistant

Distortion coefficients: [0.4373938572105337, -0.3362905676356098, 1.7958894826029195, -2.5554901038874447]

Type: aprilgrid

Tags:

Rows: 6

Cols: 6

Size: 0.021 [m]

Spacing 0.0063 [m]

IMU configuration

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IMU0:

Model: scale-misalignment

Update rate: 100.0

Accelerometer:

Noise density: 0.0018875980080012855

Noise density (discrete): 0.018875980080012855

Random walk: 2.553493980803048e-05

Gyroscope:

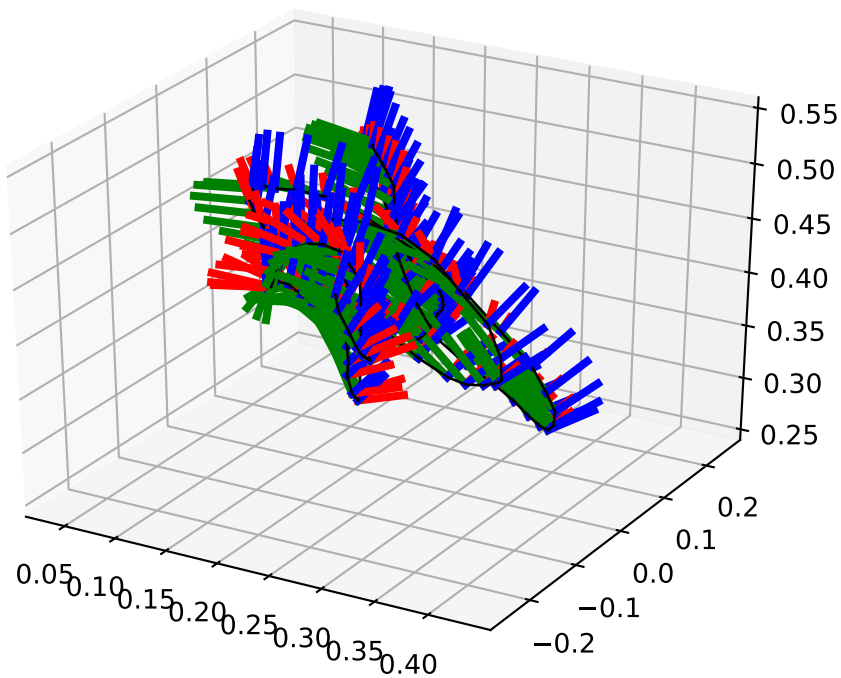
Noise density: 0.007377256400937428

Noise density (discrete): 0.07377256400937428

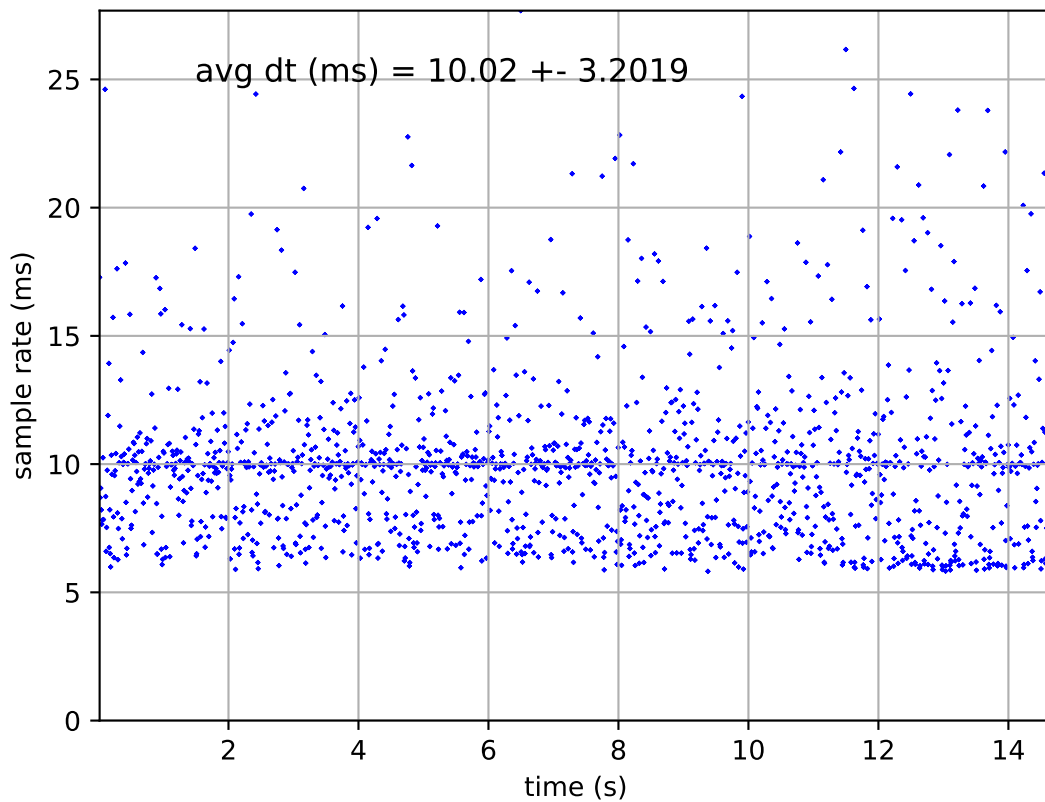
Random walk: 3.250380706253323e-05

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T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
 [0. 1. 0. 0.]
 [0. 0. 1. 0.]
 [0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]
Gyroscope:
M:
[[ 0.96054135  0.      0.      ]
 [ 0.04130229  0.9473279  0.      ]
 [-0.03265229 -0.0037676  0.98965209]]
A [(rad/s)/(m/s^2)]:
[[ 0.00888646 -0.01617145 -0.00326646]
 [ 0.00471424  0.00133884 -0.01276938]
 [-0.00355554  0.01628399 -0.0008262 ]]
C_gyro_i:
[[ 0.99844213  0.05085011 -0.02296901]
 [-0.0500746  0.99819474  0.03316303]
 [ 0.02461389 -0.0319612  0.99918599]]
Accelerometer:
M:
[[ 1.0105524  0.      0.      ]
 [-0.00489765  1.00387843  0.      ]
 [-0.00580783 -0.01509244  0.98919842]]
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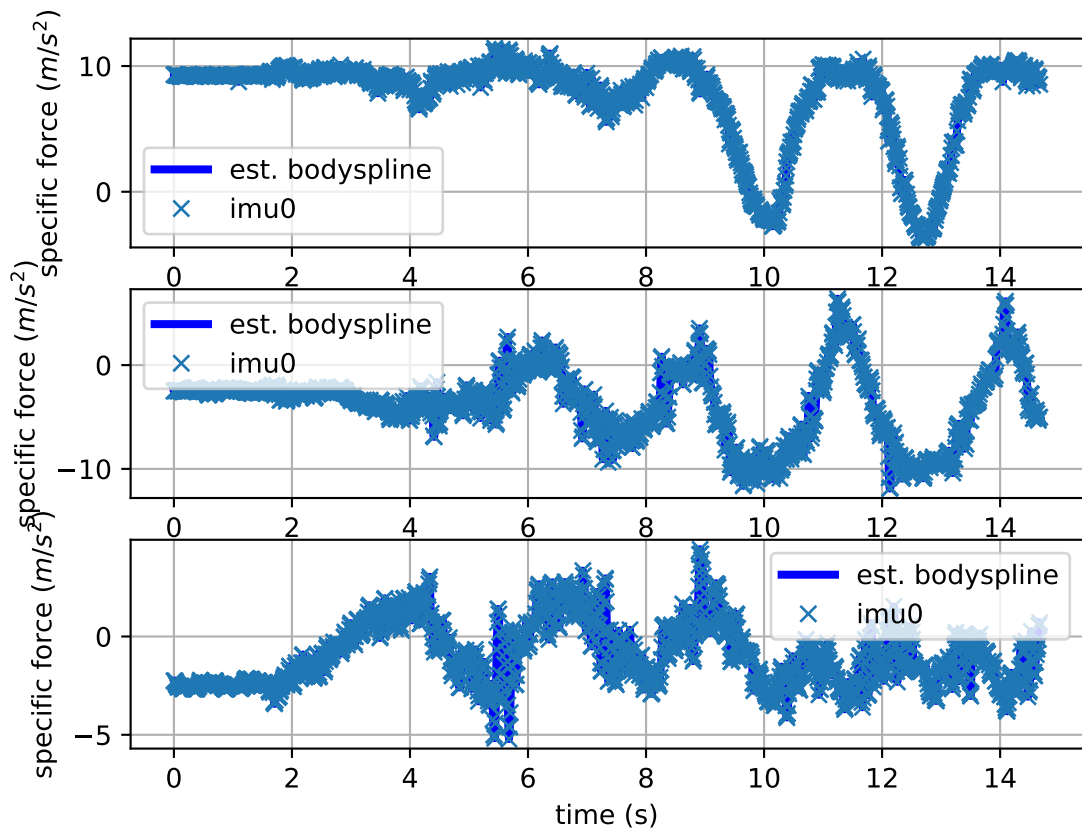
imu0: estimated poses



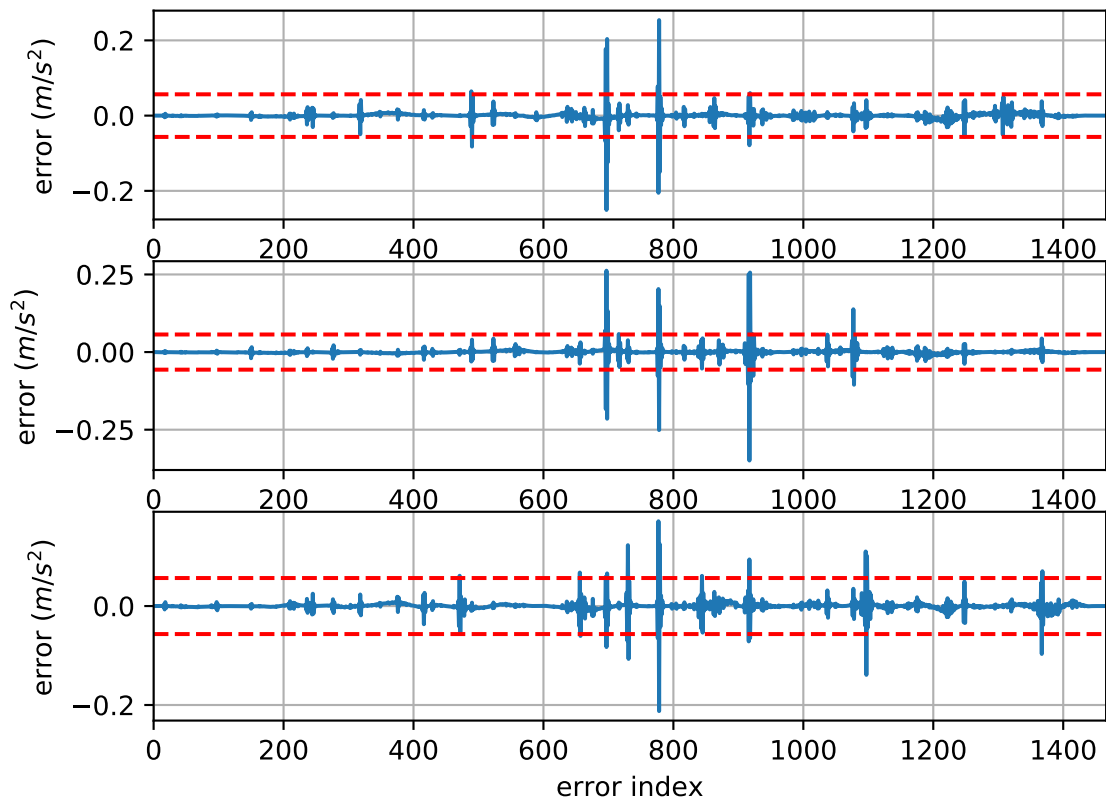
imu0: sample inertial rate



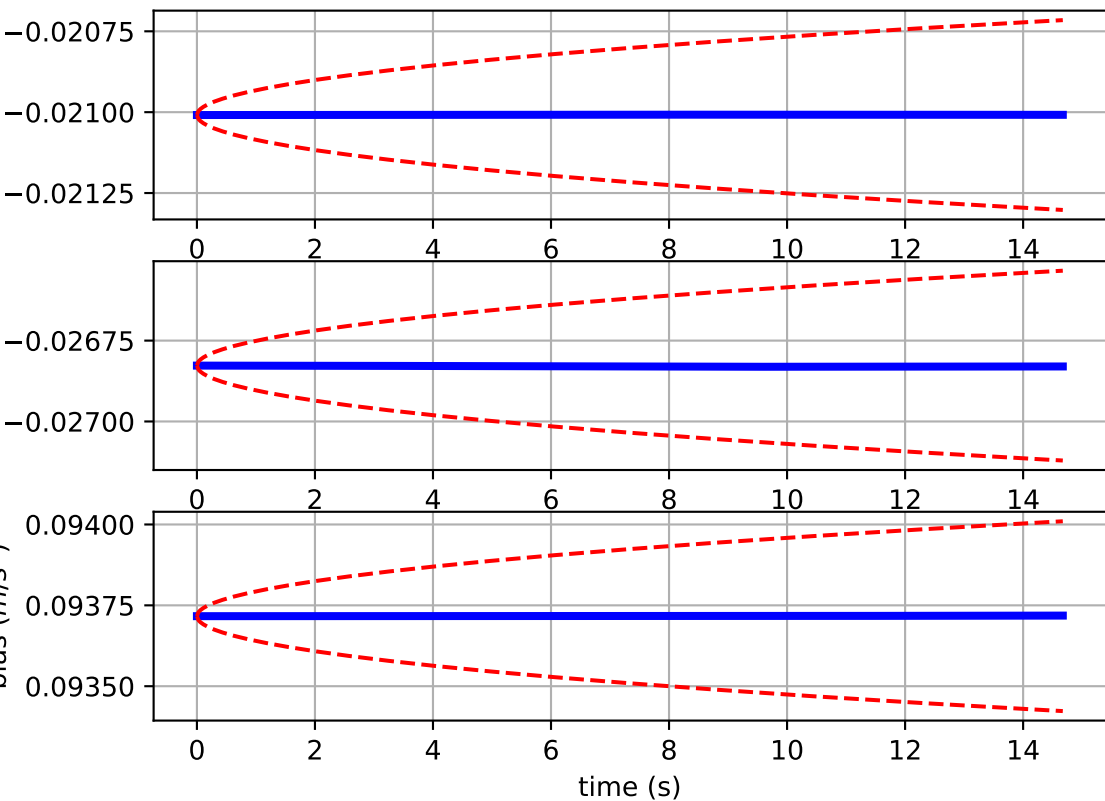
Comparison of predicted and measured specific force (imu0 frame)



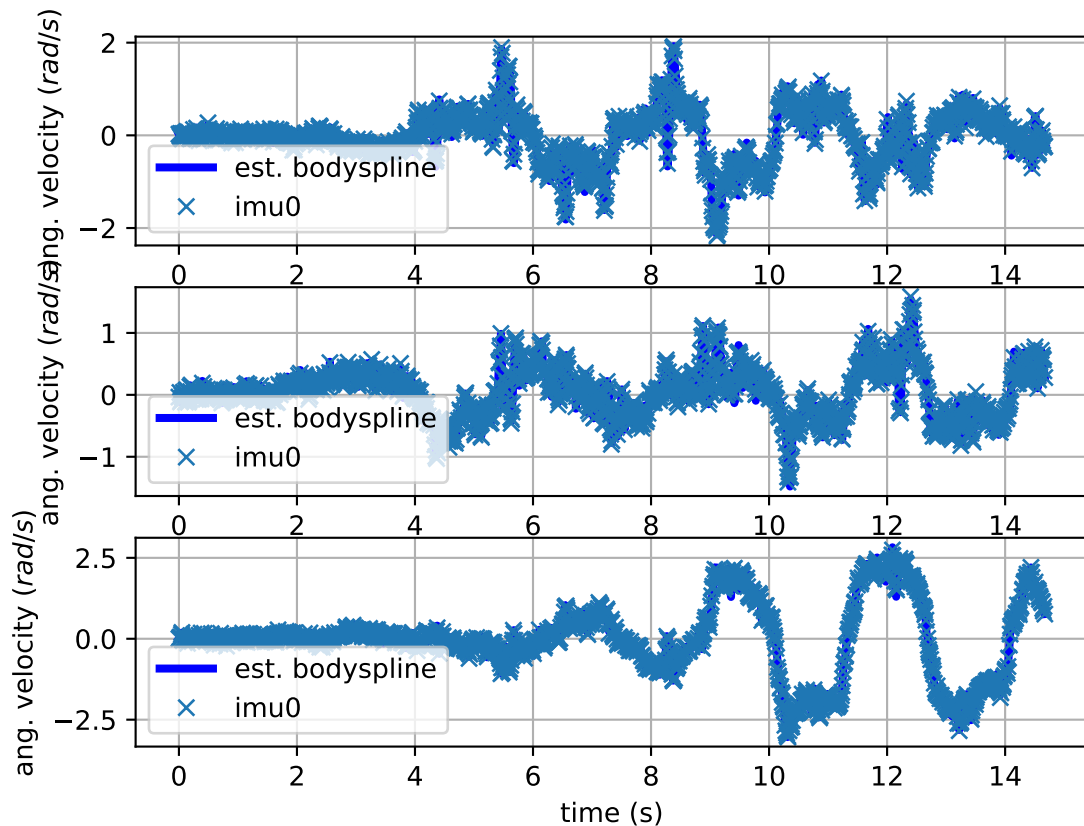
imu0: acceleration error



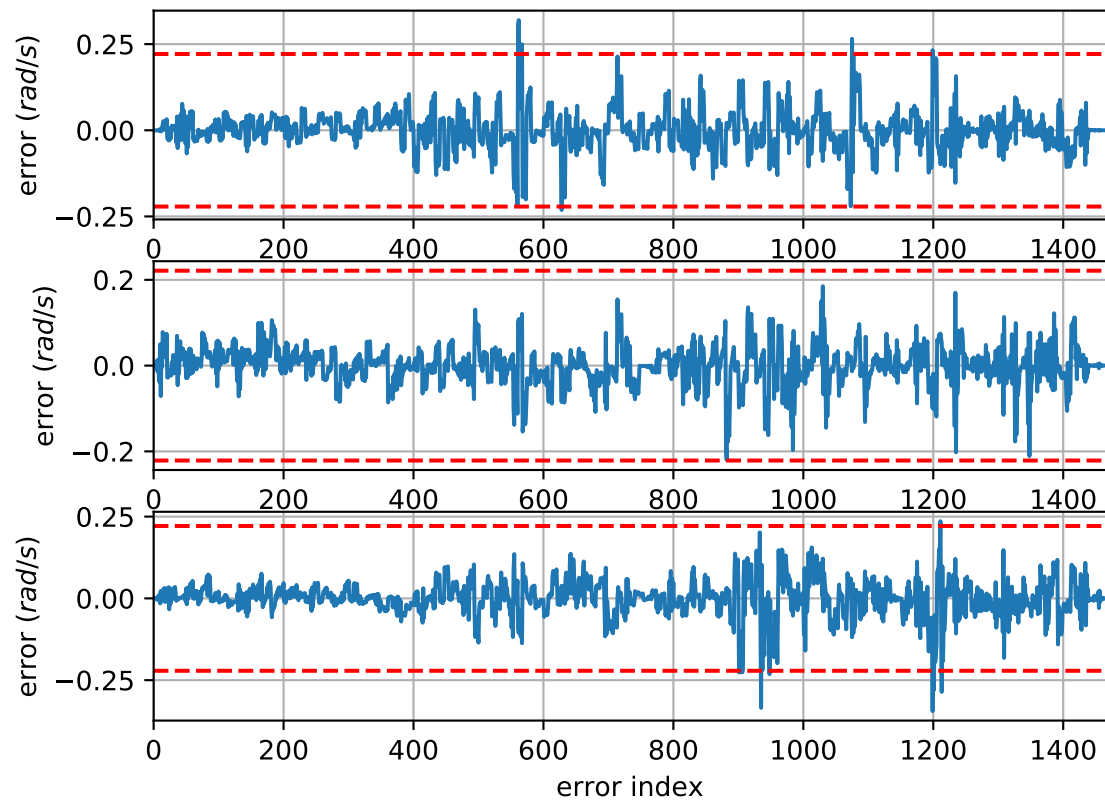
imu0: estimated accelerometer bias (imu frame)



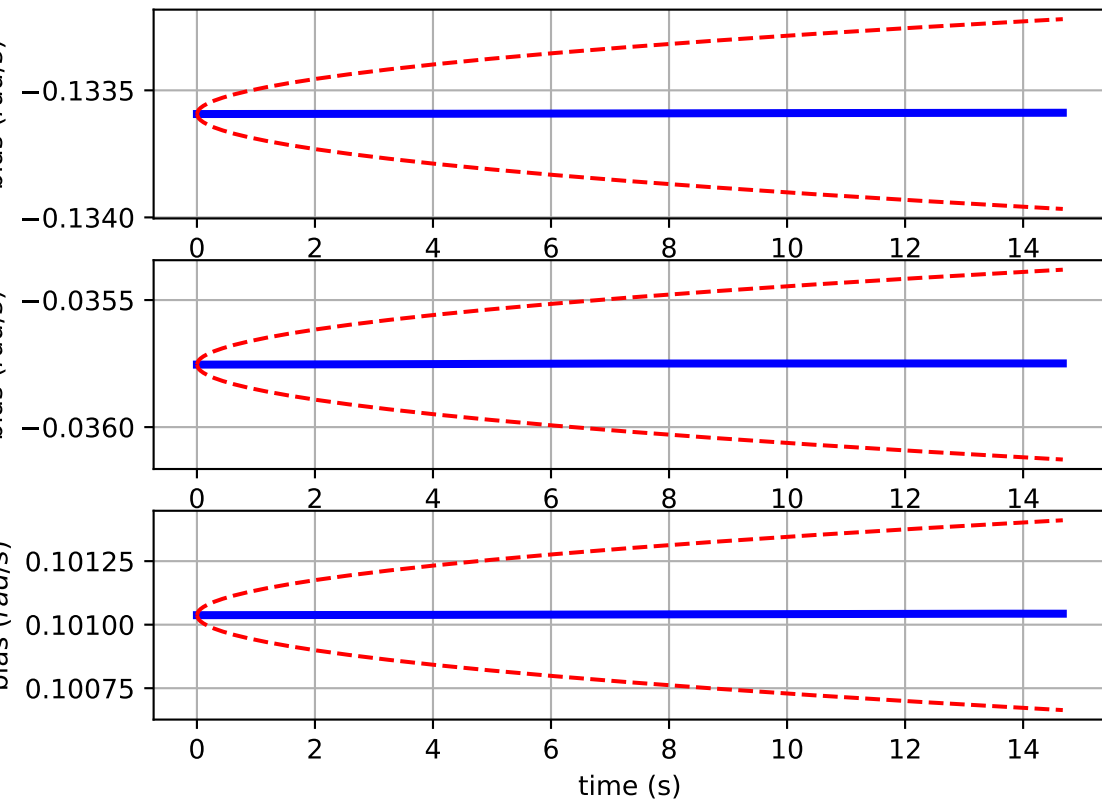
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

